

Project Validation Report

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This report summarizes the data assets in the project, baseline analytics, scenario validation, and risk analysis. Red highlights risky outcomes, while green highlights low-risk outcomes.

Project Overview

Project Metric	Value
Datasets analyzed	6
Validation scenarios	2
Baseline revenue	23,447,112.28
Baseline rows	5655

Validation Results

This section reflects the instance result produced by running the current main.py entry point, including the baseline analytics, simulation output, validation metrics, and risk score.

main.py Output	Value
Baseline total revenue	23,447,112.28
Baseline rows	5655
Simulation model	price_elasticity_linreg
Simulation revenue delta	10.10%
Simulation confidence	40.0%
Validation MAPE	3.02%
Validation directional accuracy	0.00%
Risk level	Medium
Risk score	65

Dataset Coverage

Dataset	Rows	Columns	Missing %	Notes
customer_data.csv	4500	5	0.00%	Key numeric fields: LTV, churn_flag, tenure
financials.csv	18	5	0.00%	Key numeric fields: revenue, expenses, profit
hr_data.csv	52	5	0.00%	Key numeric fields: salary
marketing_data.csv	75	5	0.00%	Key numeric fields: budget, leads, conversions
ops_data.csv	108	4	0.00%	Key numeric fields: overhead, utilization, output
sales_data.csv	5655	6	0.00%	Key numeric fields: revenue, units_sold, price_per_unit

Model Performance Metrics

Metric	Value	Threshold	Status
MAPE	3.02%	18.00%	Review
Directional Accuracy	0.00%	80.00%	Review
RMSE	23.88	-	Informational

Risk Scoring Results

Scenario	Magnitude	Predicted Revenue Delta	Confidence	Risk Level	Risk Score
price_10_pct	10 percentage	10.10%	40.0%	Medium	65
price_20_pct	20 percentage	20.21%	40.0%	High	85

price_10_pct - Medium risk (65/100): Medium risk decision because: Churn delta > 2%, Confidence < 60, Confidence < 50. Factors: Churn delta > 2%, Confidence < 60, Confidence < 50.

price_20_pct - High risk (85/100): High risk decision because: Magnitude > 10%, Churn delta > 2%, Confidence < 60, Confidence < 50. Factors: Magnitude > 10%, Churn delta > 2%, Confidence < 60, Confidence < 50.

Test Summary

Check	Result
Baseline analytics test coverage	Included
Simulation engine test coverage	Included
Risk scorer test coverage	Included
Validation metric thresholds	Review
High-risk scenarios detected	1

Performance Summary

Performance Item	Value
Report build elapsed time	0.21 seconds before PDF render
Rows processed in main sales dataset	5655
Validation scenarios evaluated	2
Simulation confidence	40.0%

Final Sign-off Status

Status: Review Required. Model thresholds need review and 1 high-risk scenarios were detected.

Threshold Reference

Model	MAPE Threshold	Directional Accuracy Threshold
Price	18.00%	80.00%

Headcount	25.00%	70.00%
Marketing	20.00%	75.00%